

Abstract

As the world has moved increasingly towards a virtual economy, a growing number of customers perform their financial services online. While most customers apply for online banking in response to the evolving landscape, some customers submit applications as results of promotion issued by the bank, but do not continue to utilize the online functionality. This results in the bank requiring larger data storage capacity to handle accounts which are not used.

In this study, we identify the characteristic of customers likely to make use online banking for lengthy timespans with a [modified mixture cure model](#), which incorporates both the willingness and the timespan aspects to estimate the customers' preference and longevity, with regards to online banking use. Except for logistic regression, the willingness is estimated using the debiased least absolute shrinkage and selection operator (debiased LASSO) method and random forest, solving the multicollinearity problem of covariates with the advantage of non-generalized linear forms. The timespan of engagement is addressed with [Weibull distribution](#).

The method is applied to a real-world case drawn from an anonymous bank in Taiwan. The results show that age, owning a securities account and residential area are significant variables, and these variables are associated with customers' preference and timespan of using online banking via the proposed models. The findings corroborate financial market trends in Taiwan.

Keywords: Online banking; Debiased LASSO model; Random forest; Mixture use model; Willingness of using online banking

Chapter 1

Mixture debiased LASSO use model and mixture random forest use model of online banking

1.1 Introduction

Since its advent in 1969, the internet has evolved from being a tool only used in academic or technical fields to becoming a mainstream channel of communication (Nehmzow, 1997). In 1995, the United States established the first online bank – Security First Network Bank (SFNB) – closely followed by online banks in Europe and Asia in 1999 and 2000, respectively. Online banking necessitates a change in the types of customer and solves many inconveniences of traditional finance. It also increases the frequency of customers' transactions in the financial market, and has resulted in a shift towards e-commerce (Crede, 1995; Ooi, 1999).

In the last two decades, online banking accounts have become increasingly popular, and as technology has evolved, so too has the banking industry. Chen et al. (2002) summarized the advantages of developing and promoting online banking services in Taiwan: (1) The internal cost is low. As an e-commerce medium, online banking does not require physical tellers or business premises, greatly reducing operating costs. (2) They provide customers with more complete and efficient services. Services in traditional banking businesses were scattered across different departments which made handling efficiency low. Online banking can provide integrated service. (3) They increase the communication and marketing channels between the bank and its customers. Customers can give feedback when they have any questions or comments. (4) Database integration and application technology differs from traditional banks. The key feature of online banking is that it can use the customer's database

to analyze the characteristics of the customer. It provides personalized financial derivative products and wealth management products. (5) The diversified functions, personalized content and the integration of financial services can increase online banking functions and create a customer-specific financial environment. (6) Online banking comes with the benefit from diversified access channels. Online banking is no longer limited to physical entities and can be used on personal computers, smart phones, or mobile devices. Moreover, an operation is not restricted by location and time.

Lassar et al. (2005) proposed the existence of relationships between consumer innovativeness, attitudes, and web usage efficacy on the internet and online banking, that highlight the close association between customers and the internet. The usage efficacy includes about web usage intensity, web usage length, and technology comfort. This was reinforced by Mavri and Ioannou (2006) who described the influence of factors, including use of internet technology, upon consumer attitudes. Previously, researchers have used questionnaire surveys to collect data exploring consumer characteristics, to identify factors that underlie preference for online banking usage. Younger age has repeatedly been identified as a customer characteristic associated with online banking usage. Calisir and Gumussoy (2008) surveyed four age groups: 18-25, 26-40, 41-55, and 56-65. The result showed that over 44% academics were using e-banking for the group of age 26-44. Kumari (2016) designed a structural questionnaire to explore online banking use for two hundred academic faculty at the University of Kelaniya in Sri Lanka, and showed a growing trend of online banking use among young academics. Socioeconomic status has also been implicated. Annin et al. (2014) used logistic regression on survey data and found that E-banking use was influenced by socioeconomic status (operational bank, occupational status, and monthly income level) in Kumasi Metropolis, Ghana. While these studies analyzed a customers perceived engagement with online banking at one snapshot in time, Duasa et al. (2019) sought to understand the probability of using digital banking services among consumers in Malaysia. They analyzed survey data by the logistic regression and found a higher probability of choosing online banking services among those who were highly educated, females, and those not working in the government sector.

However, few studies have explored the willingness of using online banking and their timespan of use concurrently, when willingness is an unobservable variable. Nonetheless, it is an important question, as some customers may open their accounts in order to take advantage of a promotion or may only use online banking for a single purpose. These customers may reduce their interactions with the online banking system over time, rendering their account obsolete. This in turn can impact the data storage capacity and the decision in the bank's managements. Dealing with the willingness of customers to use online banking and the timespan of the engagement may require a similar approach as the censoring issues

of medical care, such that the default events may not occur within the observation period, and the default is a hidden variable. Mo and Yau (2010) proposed a mixture cure model to estimate the default of personal loans. Tong et al. (2012) used a mixture cure model to estimate the default ratio and timespan of the default. Wolter and Rösch (2014) used a mixture cure model on a company's credit risk to explore the default likelihood of small and medium enterprises.

Similar to the mixture cure model, this study proposes an integrated model to identify the variables that influence the willingness and the timespan of online banking and to estimate the median of time interval from account application to the last use of online banking by the Weibull distribution. Willingness is determined via a debiased Least Absolute Shrinkage and Selection Operator (debiased LASSO) and a random forest method of non-generalized linear form to solve the multicollinearity problem of covariates, corresponding to the parametric and non-parametric methods, respectively.

This chapter is organized as follows. In section 2, we propose the modified debiased LASSO-Weibull model and the modified random forest-Weibull model. In section 3, a statistical simulation is conducted to compute estimates for the parameters of these proposed models, and their standard deviations. In section 4, we use real banking data to illustrate the statistical analysis and results. In section 5, the conclusions of this research, research limitations, and the direction of the improvements in future research are discussed.

1.2 Modified debiased LASSO-Weibull model and modified random forest Weibull model

Consider the survival function $S(t) = P(T > t)$ is by incidence part and latency parts for the concept of Yang and Ma (2017), that is

$$S(t) = (1 - \pi_u)S_h(t) + \pi_u S_u(t). \quad (1.1)$$

Here T is the random timespan. t is any given time. π_u is the probability of a non-cured state. $S_u(t)$ is the survival function under the non-cured state. $S_h(t)$ is the survival function of cured subjects and is assumed to follow an exponential distribution. At the same time, it considers whether the patient is cured (cure) and survival time (timespan) in the non-cured condition. The survival function in the cured state is set to 1. Because the cured status of patient is unknown, the status becomes a latent variable. In the current study, the concept of disease being cured has been replaced with the presence of willingness to use online banking. Additionally, the patient's survival time has been substituted with the

duration of online banking usage. Consequently, the model employed in this study is referred to as a mixture logistic regression usage model.

Assume $\tilde{T}_i$ is the random timespan from the application to the last time of using online banking for customer i before the research end. The censored time C_i is the time of customer i from the application of online banking to the research end; $\tilde{T}_i$ and C_i are independent; the censoring time is non-informative. Let $T_i = \min(\tilde{T}_i, C_i)$, and t_i be its observed value. Assume $T_1, \dots, T_n$ are independent.

Consider U_i is an unobserved variable whether customer i has the willingness of using online banking. $U_i = 1$ means that customer i is willing to use online banking; otherwise, $U_i = 0$. However, the first use time of online banking may not be within the research period. δ_i is defined as whether online banking has been used or not during the research period; $\delta_i = 1$ means that the customer i has used online banking during the research period; otherwise, $\delta_i = 0$, and it is also called the censored case. In the mixture cure model, (u_i, δ_i) only have three statuses: $(1, 1)$, $(1, 0)$ and $(0, 0)$. Because it is unreasonable for customers of no willingness but to use online banking before the research ends, $(0, 1)$ is impossible to occur. Assume $\mathbf{X}_i$ and $\mathbf{Z}_i$ are vectors of explanatory variables of customer i that affect her/his use time and the willingness of using online banking, respectively. Let $\tilde{\mathcal{O}}_i = \{t_i, \delta_i, \text{observed } U_i, \mathbf{X}_i, \mathbf{Z}_i\}$ be observed data of individual i . Assume that $\tilde{\mathcal{O}} = \{\tilde{\mathcal{O}}_i, i = 1, \dots, n\}$ is the observed data set, and $\tilde{\mathcal{C}} = \{t_i, \delta_i, U_i, \mathbf{X}_i, \mathbf{Z}_i, i = 1, \dots, n\}$ is the complete data set. Let parameters set $\boldsymbol{\vartheta} = (\boldsymbol{\gamma}^T, \boldsymbol{\beta}^T, \alpha)$, where α is the shape parameter of the baseline risk function; $\boldsymbol{\beta}$ is the parameter vector that affects the use time of online banking; $\boldsymbol{\gamma}$ is the parameter vector that affects the preference of online banking.

1.2.1 Modified logistic-Weibull model: original type

Let U_i follow a Bernoulli distribution $Ber(\pi(\mathbf{Z}_i))$, and

$$\pi(\mathbf{Z}_i) = P(U_i = 1 | \mathbf{Z}_i) = \frac{\exp(\boldsymbol{\gamma}^T \tilde{\mathbf{Z}}_i)}{1 + \exp(\boldsymbol{\gamma}^T \tilde{\mathbf{Z}}_i)},$$

where $\tilde{\mathbf{Z}}_i = (1, \mathbf{Z}_i^T)^T$. It was indicating that the probability of preference of using online banking, that is the probability of which customer i has the willingness of using online banking. Let T_i follow a Weibull distribution, with its hazard function and probability density function (pdf) $f(t_i | \mathbf{X}_i)$ is then given by

$$\begin{aligned} S_u(t_i | \mathbf{X}_i) &= \exp \left[-\exp(\boldsymbol{\beta}^T \mathbf{X}_i) t_i^\alpha \right], \\ h(t_i | \mathbf{X}_i) &= \alpha t_i^{\alpha-1} \exp(\boldsymbol{\beta}^T \mathbf{X}_i), \\ f(t_i | \mathbf{X}_i) &= h(t_i | \mathbf{X}_i) S_u(t_i | \mathbf{X}_i) = \alpha t_i^{\alpha-1} \exp(\boldsymbol{\beta}^T \tilde{\mathbf{X}}_i) \exp(-\exp(\boldsymbol{\beta}^T \tilde{\mathbf{X}}_i) t_i^\alpha), \end{aligned}$$

where $\widetilde{\mathbf{X}}_i = (1, \mathbf{X}_i^T)^T$; $h(t_i|\mathbf{X}_i)$ is the risk function in medical statistics, which is the risk of using online banking for customer i in this study. The use time is referred to as timespan hereafter. The observed-data likelihood contribution of individual i is

$$\begin{aligned}\mathcal{L}_{oi}(\boldsymbol{\vartheta}) &= \pi(\mathbf{Z}_i) f(t_i|\mathbf{X}_i), \text{ for } \delta_i = 1, \\ \mathcal{L}_{oi}(\boldsymbol{\vartheta}) &= \pi(\mathbf{Z}_i) S(t_i|\mathbf{X}_i) + (1 - \pi(\mathbf{Z}_i)) S_h(t_i), \text{ for } \delta_i = 0,\end{aligned}$$

which can be expressed as

$$\mathcal{L}_{oi}(\boldsymbol{\vartheta}) = [\pi(\mathbf{Z}_i) f(t_i|\mathbf{X}_i)]^{\delta_i} [\pi(\mathbf{Z}_i) S(t_i|\mathbf{X}_i) + (1 - \pi(\mathbf{Z}_i)) S_h(t_i)]^{1-\delta_i}$$

Assume all observations are independent, and the observed-data likelihood for the modified logistic-Weibull model is then given by

$$\begin{aligned}\mathcal{L}_o(\boldsymbol{\vartheta}) &= \prod_{i=1}^n \mathcal{L}_{oi}(\boldsymbol{\vartheta}) \\ &= \prod_{i=1}^n [\pi(\mathbf{Z}_i) f(t_i|\mathbf{X}_i)]^{\delta_i} [\pi(\mathbf{Z}_i) S(t_i|\mathbf{X}_i) + (1 - \pi(\mathbf{Z}_i)) S_h(t_i)]^{1-\delta_i}\end{aligned}\quad (1.2)$$

The complete-data likelihood contribution of individual i can be expressed as

$$\begin{aligned}\mathcal{L}_{ci}(\boldsymbol{\vartheta}) &= \pi(\mathbf{Z}_i) [f(t_i|\mathbf{X}_i)]^{\delta_i} [S(t_i|\mathbf{X}_i)]^{1-\delta_i}, \text{ for } U_i = 1, \\ \mathcal{L}_{ci}(\boldsymbol{\vartheta}) &= (1 - \pi(\mathbf{Z}_i)) S_h(t_i), \text{ for } U_i = 0,\end{aligned}$$

Thus, the complete-data likelihood contribution of individual i can be written as

$$\mathcal{L}_{ci}(\boldsymbol{\vartheta}) = \left\{ \pi(\mathbf{Z}_i) [h(t_i|\mathbf{X}_i)]^{\delta_i} [S(t_i|\mathbf{X}_i)] \right\}^{U_i} \{(1 - \pi(\mathbf{Z}_i)) S_h(t_i)\}^{1-U_i}.$$

The complete-data log-likelihood for the modified logistic-Weibull model is given by

$$\begin{aligned}\mathcal{L}_c(\boldsymbol{\vartheta}) &= \prod_{i=1}^n \mathcal{L}_{ci}(\boldsymbol{\vartheta}) \\ &= \prod_{i=1}^n \left\{ \pi(\mathbf{Z}_i) [h(t_i|\mathbf{X}_i)]^{\delta_i} [S(t_i|\mathbf{X}_i)] \right\}^{U_i} \{(1 - \pi(\mathbf{Z}_i)) S_h(t_i)\}^{1-U_i} \\ &= \prod_{i=1}^n [\pi(\mathbf{Z}_i)]^{U_i} [1 - \pi(\mathbf{Z}_i)]^{1-U_i} \prod_{i=1}^n \left\{ [h(t_i|\mathbf{X}_i)]^{\delta_i} S(t_i|\mathbf{X}_i) \right\}^{U_i} [S_h(t_i)]^{1-U_i} \\ &= \mathcal{L}_{c1}(\boldsymbol{\vartheta}) \mathcal{L}_{c2}(\boldsymbol{\vartheta}),\end{aligned}\quad (1.3)$$

where

$$\begin{aligned}\mathcal{L}_{c1}(\boldsymbol{\vartheta}) &= \prod_{i=1}^n [\pi(\mathbf{Z}_i)]^{U_i} [1 - \pi(\mathbf{Z}_i)]^{1-U_i} \\ \mathcal{L}_{c2}(\boldsymbol{\vartheta}) &= \prod_{i=1}^n \left\{ [h(t_i|\mathbf{X}_i)]^{\delta_i} S(t_i|\mathbf{X}_i) \right\}^{U_i} [S_h(t_i)]^{1-U_i}.\end{aligned}$$

The [complete-data log-likelihood](#) function of $\mathcal{L}_c(\boldsymbol{\vartheta})$ can be expressed as

$$\begin{aligned}\ell_c(\boldsymbol{\vartheta}) &= \ell_{c1}(\boldsymbol{\vartheta}) + \ell_{c2}(\boldsymbol{\vartheta}) \\ &= \sum_{i=1}^n U_i \log(\pi_i(\boldsymbol{\gamma})) + \sum_{i=1}^n (1 - U_i) \log(1 - \pi_i(\boldsymbol{\gamma})) \\ &\quad + \sum_{i=1}^n U_i \log [h_i^{\delta_i}(\boldsymbol{\alpha}, \boldsymbol{\beta}) S_{ui}(\boldsymbol{\alpha}, \boldsymbol{\beta})] + \sum_{i=1}^n (1 - U_i) \log [S_h(t_i)].\end{aligned}\tag{1.4}$$

Here

$$\begin{aligned}\ell_{c1}(\boldsymbol{\vartheta}) &= \log \mathcal{L}_{c1}(\boldsymbol{\vartheta}) = \sum_{i=1}^n U_i \log(\pi_i(\boldsymbol{\gamma})) + \sum_{i=1}^n (1 - U_i) \log(1 - \pi_i(\boldsymbol{\gamma})) \\ \ell_{c2}(\boldsymbol{\vartheta}) &= \log \mathcal{L}_{c2}(\boldsymbol{\vartheta}) = \sum_{i=1}^n U_i \log [h_i^{\delta_i}(\boldsymbol{\alpha}, \boldsymbol{\beta}) S_{ui}(\boldsymbol{\alpha}, \boldsymbol{\beta})] + \sum_{i=1}^n (1 - U_i) \log [S_h(t_i)],\end{aligned}$$

where n is the sample size, that is the number of customers, the symbols $h(t_i|\mathbf{X}_i)$, $S_u(t_i|\mathbf{X}_i)$ and $\pi(\mathbf{Z}_i)$ can be rewritten as $h_i(\boldsymbol{\alpha}, \boldsymbol{\beta})$, $S_{ui}(\boldsymbol{\alpha}, \boldsymbol{\beta})$ and $\pi_i(\boldsymbol{\gamma})$. Because u_i is an unobservable variable, I propose the expectation generalized method of moments algorithm with the logistic model (EGMM-logistic algorithm). The E-step is to take the expectation of $\ell_c(\boldsymbol{\vartheta})$ with respect to the distribution of unobserved variable given the observed data set $\tilde{\boldsymbol{\Theta}}$ under the current estimates $\hat{\boldsymbol{\theta}}$. That is

$$\begin{aligned}\tilde{\ell}_c(\boldsymbol{\vartheta}; \boldsymbol{\vartheta}^{LRM(m)}) &= \mathbb{E} \left[\ell_c(\boldsymbol{\vartheta} | \tilde{\boldsymbol{\Theta}}; \boldsymbol{\vartheta}^{LRM(m)}) \right] \\ &= \tilde{\ell}_{c1}(\boldsymbol{\gamma}; \boldsymbol{\vartheta}^{LRM(m)}) + \tilde{\ell}_{c2}(\boldsymbol{\beta}, \boldsymbol{\alpha}; \boldsymbol{\vartheta}^{LRM(m)}).\end{aligned}$$

Here the m is the m th iteration of this algorithm used to provide $\boldsymbol{\vartheta}^{LRM(m)}$. Additionally, U_i is an unobserved variable, and it need to be estimated by

$$w_i^{LRM} = \mathbb{E}(U_i | \boldsymbol{\vartheta}^{LRM(m)}, \tilde{\boldsymbol{\Theta}}) = \delta_i + (1 - \delta_i) \frac{\pi_i(\boldsymbol{\gamma}) S_{ui}(\boldsymbol{\alpha}, \boldsymbol{\beta})}{[(1 - \pi_i(\boldsymbol{\gamma})) S_h(t_i)] + \pi_i(\boldsymbol{\gamma}) S_{ui}(\boldsymbol{\alpha}, \boldsymbol{\beta})}, \tag{1.5}$$

The GMM-step is provided in detail below. The estimates can yield the following estimating equations:

$$\begin{aligned}\frac{\partial \ell_c(\vartheta, w_i^{LRM(m)})}{\partial \gamma} &= \sum_{i=1}^n \left(w_i^{LRM(m)} - \pi(\mathbf{Z}_i) \right) \tilde{\mathbf{Z}}_i = \mathbf{0}, \\ \frac{\partial \ell_c(\vartheta, w_i^{LRM(m)})}{\partial \beta} &= \sum_{i=1}^n w_i^{LRM(m)} [\delta_i - t_i^\alpha \mu(\beta)] \tilde{\mathbf{X}}_i = \mathbf{0} \\ \frac{\partial \ell_c(\vartheta, w_i^{LRM(m)})}{\partial \alpha} &= \sum_{i=1}^n w_i^{LRM(m)} \left\{ \log(t_i) [\delta_i - t_i^\alpha \mu(\beta)] + \frac{\delta_i}{\alpha} \right\} = 0.\end{aligned}$$

The profile approach is used to update the estimate of ϑ from $\vartheta^{(m)}$. Let

$$\begin{aligned}\mathbf{g}_i \gamma(\gamma; U_i) &= [U_i - \pi_i(\mathbf{Z}_i)] \tilde{\mathbf{Z}}_i \\ \mathbf{g}_i \beta(\beta, \alpha; U_i) &= U_i [\delta_i - t_i^\alpha \mu_i(\beta)] \tilde{\mathbf{X}}_i \\ g_{i\alpha}(\beta, \alpha; U_i) &= U_i \left\{ \log(t_i) [\delta_i - t_i^\alpha \mu_i(\beta)] + \frac{\delta_i}{\alpha} \right\}\end{aligned}$$

be the estimating functions, where $\mu_i(\beta) = \exp(\beta^\top \tilde{\mathbf{X}}_i)$. The estimates can be obtained by the generalized method of moments estimator (GMM), GMM-step is:

$$\begin{aligned}\hat{\gamma}^{LRM(m+1)(1)} &= \arg \min_{\gamma} \left\{ \mathbf{g}_i \gamma(\gamma; w_i^{LRM(m)}) \right\}^\top \bar{\Lambda}_{\gamma}^{-1}(\vartheta^{LRM(m)}; w^{LRM(m)}) \left\{ \mathbf{g}_i \gamma(\gamma; w_i^{LRM(m)}) \right\}, \\ \hat{\beta}^{LRM(m+1)(1)} &= \arg \min_{\beta} \left\{ \mathbf{g}_i \beta(\beta, \alpha; w_i^{LRM(m)}) \right\}^\top \bar{\Lambda}_{\beta_1}^{-1}(\vartheta^{LRM(m)}; w^{LRM(m)}) \left\{ \mathbf{g}_i \beta(\beta, \alpha; w_i^{LRM(m)}) \right\}, \\ \hat{\alpha}^{LRM(m+1)(1)} &= \arg \min_{\alpha} \left\{ g_{i\alpha}(\beta, \alpha; w_i^{LRM(m)}) \right\}^2 \bar{\Lambda}_{\alpha_1}^{-1}(\vartheta^{LRM(m)}; w^{LRM(m)}),\end{aligned}\tag{1.6}$$

where

$$\begin{aligned}\bar{\Lambda}_{\gamma} &= \frac{1}{n} \sum_{i=1}^n \left\{ \mathbf{g}_i \gamma(\gamma; w_i^{LRM(m)}) \right\}^{\otimes 2} \\ \bar{\Lambda}_{\beta_1} &= \frac{1}{n} \sum_{i=1}^n \left\{ \mathbf{g}_i \beta(\beta, \alpha; w_i^{LRM(m)}) \right\}^{\otimes 2}, \\ \bar{\Lambda}_{\alpha_1} &= \frac{1}{n} \sum_{i=1}^n \left\{ g_{i\alpha}(\beta, \alpha; w_i^{LRM(m)}) \right\}^2.\end{aligned}$$

This procedure is iterated until convergence to obtain an updated estimate $\vartheta^{LRM(m+1)} = \left((\gamma^{LRM(m+1)})^\top, (\beta^{LRM(m+1)})^\top, \alpha^{LRM(m+1)} \right)^\top$. The gmm function in R GMM package was used to obtain the solution for minimizing objective function in GMM-step. The final estimate of ϑ can be obtained via the iteration of the E-step and GMM-step.

Although logistic regression is a usual model for cure part of mixture cure model in past most studies, other flexible models for dealing with classification problems also can be employed except for logistic regression. In order to solve the multicollinearity problem of logistic regression in the covariates and generalized linear form problem, we utilize the debiased LASSO model and random forest for willingness part.

1.2.2 Modified debiased LASSO-Weibull model

Although the logistic regression can obtain the preference of using online banking and its parameter estimates, the estimators might be unstable when some covariates are linearly dependent (that is multicollinearity). It could cause a misleading explanation of result. Ridge regression (Hoerl and Kennard, 1970) and LASSO (Tibshirani, 1996) are proposed to solve the problem of multicollinearity. Ridge regression was developed by the least sum of square error adds a sum of square parameters as penalty term, it makes the variance matrix of estimators has full rank and their estimators are unique. Another way is the LASSO method which adds a sum of absolute parameters as a penalty term. LASSO method has a different property with the ridge regression, it does not only stabilize estimators but also can select significant variables when some variables are important. The complete-data log-likelihood function of $\mathcal{L}_c(\boldsymbol{\vartheta})$ with the penalty term can be expressed as

$$\begin{aligned}\ell_c(\boldsymbol{\vartheta}) &= \ell_{c1}(\boldsymbol{\vartheta}) + \ell_{c2}(\boldsymbol{\vartheta}) + \lambda \|\boldsymbol{\gamma}\|_1, \\ &= \sum_{i=1}^n U_i \log(\pi_i(\boldsymbol{\gamma})) + \sum_{i=1}^n (1 - U_i) \log(1 - \pi_i(\boldsymbol{\gamma})) \\ &\quad + \sum_{i=1}^n U_i \log [h_i^{\delta_i}(\boldsymbol{\alpha}, \boldsymbol{\beta}) S_{ui}(\boldsymbol{\alpha}, \boldsymbol{\beta})] + \sum_{i=1}^n (1 - U_i) \log [S_h(t_i)] \\ &\quad + \lambda \|\boldsymbol{\gamma}\|_1,\end{aligned}$$

and the estimators of $\boldsymbol{\gamma}$ is by

$$\min_{\boldsymbol{\gamma}} (\ell_c(\boldsymbol{\vartheta})) = \min_{\boldsymbol{\gamma}} (\ell_{c1}(\boldsymbol{\vartheta}) + \ell_{c2}(\boldsymbol{\vartheta}) + \lambda \|\boldsymbol{\gamma}\|_1),$$

where λ is a penalty parameter, and $\|\boldsymbol{\gamma}\|_1$ is the sum of absolute value for $\boldsymbol{\gamma}$. Because LASSO estimators are the bias estimators, it needs to be corrected the bias of estimators. Van de Geer et al. (2014) proposed the debiased LASSO estimators in the linear model, and it can be written as:

$$\hat{\boldsymbol{\gamma}}^{dLASSO} = \hat{\boldsymbol{\gamma}}^{LASSO} + \hat{\boldsymbol{\Sigma}}^{-1} \boldsymbol{Z}(\mathbf{Y} - \boldsymbol{Z}^T \hat{\boldsymbol{\gamma}})/n. \quad (1.7)$$

Here $\hat{\gamma}$ is a vector of estimators by the LASSO method. $\mathbf{Z} = (\mathbf{Z}_1, \dots, \mathbf{Z}_n)$ is an $m \times n$ matrix of explanatory variables. $\mathbf{Y}$ is an $n \times 1$ continuous response vector. $\hat{\Sigma}^{-1}$ is an inverse matrix of $\hat{\Sigma} = \mathbf{Z}\mathbf{Z}^T/n$. Zhu, 2019 extended Equation (1.7) to the generalized linear model and proposed the debiased LASSO estimators:

$$\hat{\gamma}^{dLASSO} = \hat{\gamma}^{LASSO} + \hat{\Sigma}^{-1}(\hat{\gamma}^{LASSO}) \frac{\partial}{\partial \gamma} \ell(\hat{\gamma}^{LASSO}). \quad (1.8)$$

In this study, $\ell(\cdot)$ is for $\ell_c(\cdot)$, and $\hat{\Sigma}^{-1}(\hat{\gamma}^{LASSO}) = -\frac{1}{n} \frac{\partial^2}{\partial \gamma \partial \gamma^T} \ell_c(\gamma) \Big|_{\gamma=\hat{\gamma}^{LASSO}}$.

Because logistic regression is replaced by the debiased LASSO model, the mixture use model is named as mixture debiased LASSO use model. Regarding to the algorithm of the mixture debiased LASSO use model, the E-step of EGMM-dLASSO algorithm is the same as Equation (1.5),

$$\begin{aligned} w_i^{dLASSO} &= E(U_i | \delta_i, t_i, \mathbf{X}_i, \mathbf{Z}_i) \\ &= \delta_i + (1 - \delta_i) \frac{\pi_i(\gamma) S_{ui}(\alpha, \beta)}{[(1 - \pi_i(\gamma)) S_h(t_i)] + \pi_i(\gamma) S_{ui}(\alpha, \beta)} \Big|_{\vartheta} = \vartheta^{dLASSO(m)}. \end{aligned}$$

$\hat{\gamma}^{dLASSO}$ is obtained by dLASSO estimators, while $\hat{\beta}$ and $\hat{\alpha}$ can be obtained by GMM estimators in the profile approach. Let

$$\begin{aligned} g_{\beta} &= w_i^{dLASSO} [\delta_i - t_i^{\alpha} \mu_i(\beta)] \mathbf{X}_i \\ g_{\alpha} &= w_{i2} \left\{ \log(t_i) [\delta_i - t_i^{\alpha} \mu_i(\beta)] + \frac{\delta_i}{\alpha} \right\} \end{aligned}$$

be the estimating functions, and the estimates be obtained by the GMM estimator:

$$\begin{aligned} \hat{\beta}_2 &= \arg \min_{\beta} \left\{ \frac{1}{n} \sum_{i=1}^n w_{i2} [\delta_i - t_i^{\alpha} \mu_i(\beta)] \mathbf{X}_i \right\}^T \bar{\Lambda}_{\beta_2}^{-1} \left\{ \frac{1}{n} \sum_{i=1}^n w_{i2} [\delta_i - t_i^{\alpha} \mu_i(\beta)] \mathbf{X}_i \right\}, \\ \hat{\alpha}_2 &= \arg \min_{\alpha} \left\{ \frac{1}{n} \sum_{i=1}^n w_{i2} \left[\log(t_i) (\delta_i - t_i^{\alpha} \mu_i(\beta)) + \frac{\delta_i}{\alpha} \right] \right\}^2 \bar{\Lambda}_{\alpha_2}^{-1}, \end{aligned} \quad (1.9)$$

where

$$\begin{aligned} \bar{\Lambda}_{\beta_2} &= \frac{1}{n} \sum_{i=1}^n \{w_{i2} [\delta_i - t_i^{\alpha} \mu_i(\beta)] \mathbf{X}_i\}^{\otimes 2}, \\ \bar{\Lambda}_{\alpha_2} &= \frac{1}{n} \sum_{i=1}^n \left\{ w_{i2} \left[\log(t_i) [\delta_i - t_i^{\alpha} \mu_i(\beta)] + \frac{\delta_i}{\alpha} \right] \right\}^2. \end{aligned}$$

On the other hand, the estimate of γ is obtained by LASSO:

$$\sum_{i=1}^n (w_i - \pi_i(\gamma)) \mathbf{Z}_i + \tau \hat{\kappa} = \mathbf{0},$$

$\hat{\boldsymbol{\kappa}} = (\hat{\kappa}_1, \dots, \hat{\kappa}_m)^\top$ is partially differentiate $\|\boldsymbol{\gamma}\|_1$ for $\boldsymbol{\gamma}$, $\hat{\kappa}_j = \text{sign}(\hat{\gamma}_j)$ of LASSO estimators if $\hat{\gamma}_j \neq 0$; τ is a penalty parameter. The unbiased estimate $\hat{\boldsymbol{\gamma}}_2$ is by Equation (1.8) on the estimate of $\boldsymbol{\gamma}$. Let $\hat{\boldsymbol{\theta}}_2 = (\hat{\boldsymbol{\gamma}}_2, \hat{\boldsymbol{\beta}}_2, \hat{\alpha}_2)$ be by EGMM-dLASSO algorithm proposed.

1.2.3 Modified random forest Weibull model

The debiased LASSO model is also a generalized linear model. In order to increase the flexibility of estimating the preference of using online banking. Li et al. (2020) proposed the mixture cure model with the support vector machine (SVM) to solve the limitation of logistic regression. However, the disadvantage of SVM is that it has not good performance when the dataset is large. More specifically, because SVM needs matrix operations for finding the hyperplane, it spends more running time for dealing with large dataset. As a result, I apply the random forest (RF) method in the willingness part, and this model is named as modified random forest Weibull model. Because the construction of RF is an nonparametric model, there is no estimate $\hat{\boldsymbol{\gamma}}$. The step is:

$$w_{i3} = E(U_i | \delta_i, t_i, \mathbf{X}_i, \mathbf{Z}_i) = \delta_i + (1 - \delta_i) \frac{\pi_i S_{ui}(\alpha, \boldsymbol{\beta})}{(1 - \pi_i) + \pi_i(\boldsymbol{\gamma}) S_{ui}(\alpha, \boldsymbol{\beta})},$$

and the estimators of $(\alpha_3, \boldsymbol{\beta}_3)$ are the same as Equation (1.9) in the GMM step.

$$\begin{aligned} \hat{\boldsymbol{\beta}}_3 &= \arg \min_{\boldsymbol{\beta}} \left\{ \frac{1}{n} \sum_{i=1}^n w_{i3} [\delta_i - t_i^\alpha \mu_i(\boldsymbol{\beta})] \mathbf{X}_i \right\}^\top \bar{\boldsymbol{\Lambda}}_{\boldsymbol{\beta}_2}^{-1} \left\{ \frac{1}{n} \sum_{i=1}^n w_{i3} [\delta_i - t_i^\alpha \mu_i(\boldsymbol{\beta})] \mathbf{X}_i \right\}, \\ \hat{\alpha}_3 &= \arg \min_{\alpha} \left\{ \frac{1}{n} \sum_{i=1}^n w_{i3} \left[\log(t_i) (\delta_i - t_i^\alpha \mu_i(\boldsymbol{\beta})) + \frac{\delta_i}{\alpha} \right] \right\}^2 \bar{\Lambda}_{\alpha_2}^{-1}, \end{aligned} \quad (1.10)$$

where

$$\begin{aligned} \bar{\boldsymbol{\Lambda}}_{\boldsymbol{\beta}_3} &= \frac{1}{n} \sum_{i=1}^n \{w_{i3} [\delta_i - t_i^\alpha \mu_i(\boldsymbol{\beta})] \mathbf{X}_i\}^{\otimes 2}, \\ \bar{\Lambda}_{\alpha_3} &= \frac{1}{n} \sum_{i=1}^n \left\{ w_{i3} \left[\log(t_i) [\delta_i - t_i^\alpha \mu_i(\boldsymbol{\beta})] + \frac{\delta_i}{\alpha} \right] \right\}^2. \end{aligned}$$

This algorithm is called [an](#) EGMM-RF algorithm.

Regarding of the estimated probabilities by RF, I propose the use of Bayesian parallel model (BPM) calibration to adjust these voting results. The BPM calibration is achieved through a linear combination of the posterior mean of voting results obtained from the random forest (RF) and the estimated probabilities from the logistic regression model (LRM). The estimated probability of π_i based on the BPM calibration are given as follows:

$$\hat{\pi}_i^{BPM} = D \hat{\pi}_i^{BRF} + (1 - D) \hat{\pi}_i^{LRM}. \quad (1.11)$$

Here $\hat{\pi}_i^{BRF}$ is the posterior means of voting results for the random forest, the estimated probabilities $\hat{\pi}_i^{LRM}$ is by the predicted probabilities for the logistic regression model, and the default weighting value D is 1/2. Assume the prior distribution of $(\pi_i, 1 - \pi_i)$ follows a Beta distribution with two parameters ζ_1 and ζ_2 . The pdf of the Beta distribution is defined as follows:

$$f(\pi_i, 1 - \pi_i) = \frac{\Gamma(\zeta_1 + \zeta_2)}{\Gamma(\zeta_1)\Gamma(\zeta_2)} \pi_i^{\zeta_1-1} (1 - \pi_i)^{\zeta_2-1},$$

where $\Gamma(\zeta_k) = (\zeta - 1)!, k = 1, 2$. According to the above assumption, the posterior means π_i^{BRF} is given as follows:

$$\pi_i^{BRF} = \frac{1}{1 + \zeta_1 + \zeta_2} U_i + \frac{\zeta_1}{1 + \zeta_1 + \zeta_2}.$$

Here because U_i is an unobserved variable, it can be replaced by the voting results of RF. Regarding to the setting of ζ_1 and ζ_2 , the non-informative method is that ζ_1 and ζ_2 are set to 1. The maximum likelihood estimates of ζ_1 and ζ_2 can be used as ζ_1 and ζ_2 , these estimates are obtained by maximizing the likelihood function

$$\mathcal{L}(\zeta_1, \zeta_2) = \prod_{i=1}^n f(\hat{\pi}_i^{LRM}, 1 - \hat{\pi}_i^{LRM}) = \prod_{i=1}^n \left[\frac{\Gamma(\zeta_1 + \zeta_2)}{\Gamma(\zeta_1)\Gamma(\zeta_2)} (\hat{\pi}_i^{LRM})^{\zeta_1-1} (1 - \hat{\pi}_i^{LRM})^{\zeta_2-1} \right].$$

This algorithm is named as EGMM-RF algorithm, and the proposed estimation procedure involves the following E-step and GMM-step:

- Step 1. Use the censored status δ_i as [an initial value of](#) $w_{i3}^{BPM(0)}$.
- Step 2. Use $w_{i3}^{BPM(m)}$ and the procedure as given in Equation to update estimates of α and β , denoted as $\alpha^{BPM(m+1)}$ and $\beta^{BPM(m+1)}$.
- Step 3. This step begins with the imputation of u_i by multiple imputation. Assume u_i is generated from $U_i \sim \text{Bernoulli distribution } Ber(w_{i3}^{BPM(m)})$ to obtain estimated probability of π_i by BRF calibration, and obtain estimated probability of π_i via LRM calibration using the linear combination in each imputed data set. The average of these repeated estimated probability by using the BPM calibration serve as estimates of π_i , for patient i , denoted as $\pi_i^{BPM(m+1)}$. This entire procedure involves 5 multiple imputations.
- Step 4. Update w_{i3} denoted as $w_{i3}^{BPM(m+1)}$, by $\pi_{i3}^{BPM(m+1)}$, $\alpha^{BPM(m+1)}$, and $\beta^{BPM(m+1)}$.
- Step 5. Repeat [Steps 2, 3, and 4 until convergence](#).

Let $\hat{\theta}_3 = (\hat{\beta}_3, \hat{\alpha}_3)$ be by EGMM-RF algorithm

1.2.4 Variances of estimators of parameters

The sandwich variance estimators are used to estimate the variances of estimators obtained through the EGMM-logistic algorithm and the EGMM-dLASSO algorithm. For the EGMM-RF algorithm, on the other hand, bootstrap variance estimators are used to estimate the variances of its estimators.

Given the estimator of parameter α as an example of bootstrap variance estimators, other parameters follow the same procedure. Let $\hat{\alpha}_b$ be the estimator of α from the b th bootstrapping sample, $b = 1, \dots, B$, B is the number of bootstrapping samples. The bootstrap estimated variance of $\hat{\alpha}$ is:

$$\widehat{\text{Var}}(\hat{\alpha}) = \sum_{b=1}^B (\hat{\alpha}_b - \bar{\alpha})^2 / (B - 1)$$

where $\bar{\alpha} = \sum_{b=1}^B \hat{\alpha}_b / B$. According to the central limit theorem, when the sample size is large, the $(1 - a)\%$ confidence interval of α is $(\hat{\alpha} - z_{\frac{a}{2}} \sqrt{\widehat{\text{Var}}(\hat{\alpha})}, \hat{\alpha} + z_{\frac{a}{2}} \sqrt{\widehat{\text{Var}}(\hat{\alpha})})$.

If the sample size is small, the upper confidence limit and the lower confidence limit of α could be built by the $100(a/2)\%$ percentile and $100(1 - a/2)\%$ percentile from the bootstrapping samples.

On the other hand, the variances of estimators obtained through the EGMM-logistic algorithm and EGMM-dLASSO can be utilized for sandwich variance estimator. Let

$$\mathbf{q}(\boldsymbol{\theta}) = \left([U_i - \pi_i(\boldsymbol{\gamma})] \mathbf{Z}_i^T, U_i [\delta_i - t_i^\alpha \mu_i(\boldsymbol{\beta})] \mathbf{X}_i^T, U_i \left\{ \log(t_i) [\delta_i - t_i^\alpha \mu_i(\boldsymbol{\beta})] + \frac{\delta_i}{\alpha} \right\} \right)^T,$$

and the sandwich variance estimator be:

$$\boldsymbol{\Sigma} = n^{-1} \mathbf{A}_o^{-1}(\boldsymbol{\theta}) \mathbf{V}_o(\boldsymbol{\theta}) \mathbf{A}_o^{-1}(\boldsymbol{\theta}), \quad (1.12)$$

where $\mathbf{A}_o(\boldsymbol{\theta})$ is $\text{E} \left[-\frac{\partial}{\partial \boldsymbol{\theta}} \mathbf{q}(\boldsymbol{\theta}) \right] - \mathbf{A}_m(\boldsymbol{\theta})$, and $\mathbf{V}(\boldsymbol{\theta}) = \text{E} [\mathbf{q}^{\otimes 2}(\boldsymbol{\theta})]$. Here

$$\begin{aligned} & \text{E} \left[-\frac{\partial}{\partial \boldsymbol{\theta}} \mathbf{q}(\boldsymbol{\theta}) \right] \\ &= \text{E} \begin{pmatrix} (\pi_i(\boldsymbol{\gamma}) - \pi_i^2(\boldsymbol{\gamma})) \mathbf{Z}_i^{\otimes 2} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & u_i t_i^\alpha \mu_i(\boldsymbol{\beta}) \mathbf{X}_i^{\otimes 2} & u_i \log(t_i) t_i^\alpha \mu_i(\boldsymbol{\beta}) \mathbf{X}_i^T \\ \mathbf{0} & u_i t_i^\alpha \log(t_i) \mu_i(\boldsymbol{\beta}) & u_i t_i^\alpha (\log(t_i))^2 \mu_i(\boldsymbol{\beta}) + \frac{\delta_i}{\alpha^2} \end{pmatrix}, \\ & \mathbf{A}_m(\boldsymbol{\theta}) \\ &= \text{E} \left((U_i - w_i^2) \begin{pmatrix} \mathbf{Z}_i^{\otimes 2} & \mathbf{B}_{12i} & \mathbf{B}_{13i} \\ \mathbf{B}_{12i}^T & [\delta_i - t_i^\alpha \mu_i(\boldsymbol{\beta})] \mathbf{X}_i^{\otimes 2} & \mathbf{B}_{23i} \\ \mathbf{B}_{13i}^T & \mathbf{B}_{23i}^T & \{\log(t_i)(\delta_i - t_i^\alpha \mu_i(\boldsymbol{\beta}) + \frac{\delta_i}{\alpha})\}^2 \end{pmatrix} \middle| \delta_i, t_i, \mathbf{X}_i, \mathbf{Z}_i \right), \end{aligned}$$

and the matrix $\mathbf{V}(\boldsymbol{\theta}) = \mathbb{E}[\mathbf{q}(\boldsymbol{\theta})^{\otimes 2}]$. $\widehat{\boldsymbol{\Sigma}}$ can be obtained by substituting $\boldsymbol{\theta}$ with $\widehat{\boldsymbol{\theta}}_k = (\widehat{\gamma}_k, \widehat{\boldsymbol{\beta}}_k, \widehat{\alpha}_k)$, $k = 1, 2$; U_i can be replaced by w_i . $\mathbb{E}[(U_i - w_i)^2 | c_i, t_i, \mathbf{X}_i, \mathbf{Z}_i] = w_i(1 - w_i)$.

$$\widehat{\boldsymbol{\Sigma}} = n^{-1} \overline{\mathbf{A}}^{-1}(\widehat{\boldsymbol{\theta}}) \overline{\mathbf{V}}(\widehat{\boldsymbol{\theta}}) \overline{\mathbf{A}}^{-1}(\widehat{\boldsymbol{\theta}}),$$

Here

$$\begin{aligned} \overline{\mathbf{A}}(\widehat{\boldsymbol{\theta}}_k) &= \frac{1}{n} \sum_{i=1}^n \left[-\frac{\partial}{\partial \boldsymbol{\theta}} \mathbf{q}(\boldsymbol{\theta}) \Big|_{\boldsymbol{\theta}=\widehat{\boldsymbol{\theta}}_k} \right] - \overline{\mathbf{A}}_m(\widehat{\boldsymbol{\theta}}_k), \\ \overline{\mathbf{V}}(\widehat{\boldsymbol{\theta}}_k) &= \frac{1}{n} \sum_{i=1}^n \mathbf{q}(\boldsymbol{\theta})^{\otimes 2} \Big|_{\boldsymbol{\theta}=\widehat{\boldsymbol{\theta}}_k}, k = 1, 2, \end{aligned}$$

and

$$\begin{aligned} &\overline{\mathbf{A}}_m(\boldsymbol{\theta}) \\ &= \frac{1}{n} \sum_{i=1}^n \left(w_i(1 - w_i) \begin{pmatrix} \mathbf{Z}_i \mathbf{Z}_i^{\top} & \mathbf{B}_{12i} & \mathbf{B}_{13i} \\ \mathbf{B}_{12i}^{\top} & (\delta_i - t_i^{\alpha} \mu_i(\boldsymbol{\beta}) \mathbf{X}_i^{\otimes 2} & \mathbf{B}_{23i} \\ \mathbf{B}_{13i}^{\top} & \mathbf{B}_{23i}^{\top} & (\log(t_i)(\delta_i - t_i^{\alpha} \mu_i(\boldsymbol{\beta}) + \frac{\delta_i}{\alpha}))^2 \end{pmatrix} \right), \end{aligned}$$

where

$$\begin{aligned} \mathbf{B}_{12i} &= \mathbf{Z}_i \mathbf{X}_i^{\top} [\delta_i - t_i^{\alpha} \mu_i(\boldsymbol{\beta})], \\ \mathbf{B}_{13i} &= \mathbf{Z}_i \log(t_i) \left[\delta_i - t_i^{\alpha} \mu_i(\boldsymbol{\beta}) + \frac{\delta_i}{\alpha} \right], \\ \mathbf{B}_{23i} &= \mathbf{X}_i [\delta_i - t_i^{\alpha} \mu_i(\boldsymbol{\beta})] \left[\delta_i - t_i^{\alpha} \mu_i(\boldsymbol{\beta}) + \frac{\delta_i}{\alpha} \right] \log(t_i). \end{aligned}$$

According to the robustness of sandwich variance estimator, because LASSO is nearly the logistic regression, and dLASSO estimator is unbiased at parameters of logistic regression, Equation (1.12) can also provide a robust variance for estimators of dLASSO.

1.3 Simulation study

To follow the characteristic of the real data, we consider the effect of multicollinearity with related covariates and its interaction in the estimation. Let sample size n be 300, and two sets of random variables be (X_1, X_2) and (Z_1, Z_2) . Suppose Z_2 is from Bernoulli distribution, its probability of success is 0.5. To generate a relationship with Z_1 and Z_2 , Z_1 is generated from a normal distribution which population mean is μ_0 when Z_2 is 0; Otherwise, Z_1 is generated from a normal distribution which population mean is μ_1 . The correlation of Z_1

and Z_2 is ρ . $\mu_0 = 0$ and ρ are given to adjust μ_1 . Let X_1 and X_2 be the same as Z_1 and Z_2 . Z_1 and Z_2 are covariates that affect the willingness state, while X_1 and X_2 are covariates that affect the timespan. Let u_i follow binominal distribution $Bin(1, \pi_i(\gamma))$, and the preference $\pi_i(\gamma)$ of the willingness is assumed by logistic regression,

$$\pi_i(\gamma) = \frac{\exp(\gamma_0 + \gamma_1 z_{1i} + \gamma_2 z_{2i} + \gamma_3 z_{3i})}{1 + \exp(\gamma_0 + \gamma_1 z_{1i} + \gamma_2 z_{2i} + \gamma_3 z_{3i})}$$

where z_{1i} and z_{2i} are observed values of Z_{1i} and Z_{2i} for customer i , z_{3i} is an interaction between z_{1i} and z_{2i} . The timespan of using online banking T_i is generated by PH model with a survival function equal to $\exp(-\exp(\beta_0 + \beta_1 x_1 + \beta_2 x_2)t_i^\alpha)$, and the censoring time C_i is generated independently from uniform distribution in $[0, 12]$. The simulated number of datasets is 1000. The number of growing trees for random forest is 100, and the bootstrap samples are 100 for estimating variances of estimators by EGMM-RF algorithm. Assume $S_h(t_i) = \exp(-t_i/120)$. We observe the performance of the estimation and inference of parameters under different correlation. Let correlation parameter $\rho = 0, 0.45$ be corresponding to independence, weak correlation and strong correlation as $\mu_1 = 0, 1$. In generating the simulated data, we generate C_i and U_i . When $U_i = 1$, the observed time is $\min(T_i, C_i)$ and the censoring status δ is $I(T_i \leq C_i)$. When $U_i = 0$, the observed time is C_i and the censoring status δ_i is 0.

To compare estimated preferences by the EGMM-logistic algorithm, EGMM-dLASSO algorithm, and EGMM-RF algorithm, the average mean square error (AMSE) is adopted. It is defined as:

$$\frac{1}{300} \sum_{i=1}^{300} \left(\frac{\sum_{j=1}^{1000} (\hat{\pi}_i^j - \pi_i^j)^2}{1000} \right).$$

Three algorithms are run with Intel(R) Core(TM) i7-8700.

Table 1.1 shows that the AMSEs of the three modified mixture models increase when the correlation goes to increase. As the preference of willingness follows a logit function in the simulation, it is anticipated that the AMSEs of estimators derived from the modified logistic-Weibull model and the modified debiased LASSO-Weibull model are smaller than the AMSEs of estimators obtained from the modified random forest-Weibull model. Conversely, the AMSEs of estimators produced by the modified debiased LASSO-Weibull model is expected to be lower than that of the modified logistic-Weibull model. While the disparity between the two AMSE values is subtle, the debiased LASSO model demonstrates greater suitability in capturing the nuances of the willingness part.

Table 1.2 illustrates the impact of covariates on the willingness component. Since the modified random forest-Weibull model does not need to estimate parameters $\gamma =$

$(\gamma_0, \gamma_1, \gamma_2, \gamma_3)$, the results in Table 1.2 exclusively showcase the outcomes of the modified logistic-Weibull model and the modified debiased LASSO-Weibull model. The findings indicate that the modified debiased LASSO-Weibull model surpasses the logistic-Weibull model. The bias and simulated standard error of estimators produced by the modified debiased LASSO-Weibull model are slightly lower than those of the modified logistic-Weibull model. As the correlation increases, the mean square error (MSE) of $\hat{\gamma}_2$ and $\hat{\gamma}_3$ also rises for the two models.

These two parameters represent the coefficients of the binomial covariate z_2 and the interaction term z_3 . When variables exist the relationship, adding the interaction can be stabilized in the estimation of parameters for the modified logistic-Weibull model. In cases where a linear relationship exists between z_1 and z_2 , the bias and simulated standard error for estimating the coefficient of z_2 are more pronounced than those for the continuous covariate z_1 . Even with the addition of interaction, both bias and simulated standard error experience an increase. The estimated variances closely align with simulated variances, resulting around the expected 95% coverage percentage for a confidence interval.

Tables 1.3 and 1.4 reveal that the outcomes of the modified random forest-Weibull model indicate larger biases in the estimates of the timespan compared to the other two models. This is particularly noticeable when the covariate is a binary variable.

Table 1.1: AMSEs of three methods for the preference of using online banking

Algorithm	AMSE ($\rho = 0; n = 300$)	AMSE ($\rho = 0.45; n = 300$)	AMSE ($\rho = 0; n = 500$)	AMSE ($\rho = 0.45; n = 500$)
EGMM-logistic	0.00372	0.00421	0.00218	0.00256
EGMM-dLASSO	0.00332	0.00417	0.00215	0.00254
EGMM-RFBRM	0.00659	0.00816	0.00588	0.00748

Table 1.2: The estimation of coefficients in the willingness part

$n = 300; \rho = 0$	Algorithm	Estimate	Bias	SE	SE*	MSE	CP
$\gamma_0 = 0.5$	EGMM-logistic	0.559	0.059	0.250	0.246	0.066	0.957
	EGMM-dLASSO	0.522	0.052	0.245	0.244	0.062	0.960
$\gamma_1 = 0.6$	EGMM-logistic	0.650	0.050	0.276	0.272	0.079	0.950
	EGMM-dLASSO	0.639	0.039	0.269	0.269	0.074	0.953
$\gamma_2 = 1.0$	EGMM-logistic	0.994	-0.006	0.366	0.347	0.134	0.947
	EGMM-dLASSO	0.993	-0.007	0.357	0.347	0.128	0.951
$\gamma_3 = -0.5$	EGMM-logistic	-0.543	-0.043	0.381	0.371	0.147	0.946
	EGMM-dLASSO	-0.530	-0.030	0.371	0.367	0.138	0.950
$n = 300; \rho = 0.45$							
$\gamma_0 = 0.5$	EGMM-logistic	0.563	0.063	0.251	0.248	0.067	0.958
	EGMM-dLASSO	0.561	0.061	0.250	0.248	0.066	0.957
$\gamma_1 = 0.6$	EGMM-logistic	0.655	0.055	0.275	0.275	0.078	0.957
	EGMM-dLASSO	0.651	0.051	0.273	0.274	0.077	0.957
$\gamma_2 = 1.0$	EGMM-logistic	1.006	0.006	0.433	0.423	0.187	0.964
	EGMM-dLASSO	1.000	<0.001	0.428	0.422	0.183	0.965
$\gamma_3 = -0.5$	EGMM-logistic	-0.524	-0.024	0.456	0.439	0.208	0.952
	EGMM-dLASSO	-0.522	-0.022	0.447	0.436	0.200	0.952
$n = 500; \rho = 0$							
$\gamma_0 = 0.5$	EGMM-logistic	0.544	0.044	0.187	0.187	0.037	0.947
	EGMM-dLASSO	0.541	0.041	0.186	0.186	0.036	0.948
$\gamma_1 = 0.6$	EGMM-logistic	0.646	0.046	0.204	0.206	0.044	0.964
	EGMM-dLASSO	0.640	0.040	0.202	0.205	0.042	0.965
$\gamma_2 = 1.0$	EGMM-logistic	0.991	-0.009	0.266	0.263	0.071	0.961
	EGMM-dLASSO	0.990	-0.010	0.263	0.262	0.069	0.963
$\gamma_3 = -0.5$	EGMM-logistic	-0.516	-0.016	0.281	0.282	0.079	0.945
	EGMM-dLASSO	-0.509	-0.009	0.279	0.280	0.078	0.955
$n = 500; \rho = 0.45$							
$\gamma_0 = 0.5$	EGMM-logistic	0.546	0.046	0.188	0.188	0.037	0.948
	EGMM-dLASSO	0.546	0.046	0.188	0.188	0.037	0.948
$\gamma_1 = 0.6$	EGMM-logistic	0.649	0.049	0.205	0.208	0.044	0.963
	EGMM-dLASSO	0.647	0.047	0.205	0.208	0.044	0.963
$\gamma_2 = 1.0$	EGMM-logistic	0.983	-0.017	0.330	0.319	0.109	0.948
	EGMM-dLASSO	0.981	-0.019	0.329	0.318	0.108	0.948
$\gamma_3 = -0.5$	EGMM-logistic	-0.509	-0.009	0.322	0.325	0.103	0.955
	EGMM-dLASSO	-0.507	-0.007	0.320	0.325	0.102	0.955

Table 1.3: The estimation of coefficients in timespan part

$n = 300; \rho = 0$	Algorithm	Estimate	Bias	SE	SE*	MSE	CP
$\alpha = 1.0$	EGMM-logistic	1.014	0.014	0.059	0.060	0.004	0.945
	EGMM-dLASSO	1.015	0.015	0.059	0.060	0.004	0.945
	EGMM-RF	1.019	0.019	0.058	0.061	0.004	0.949
$\beta_0 = -0.3$	EGMM-logistic	-0.310	-0.010	0.146	0.141	0.021	0.950
	EGMM-dLASSO	-0.308	-0.008	0.146	0.141	0.021	0.951
	EGMM-RF	-0.294	0.006	0.141	0.140	0.020	0.953
$\beta_1 = -1.0$	EGMM-logistic	-1.019	-0.019	0.099	0.098	0.010	0.947
	EGMM-dLASSO	-1.019	-0.019	0.099	0.098	0.010	0.947
	EGMM-RF	-0.991	0.009	0.097	0.100	0.009	0.953
$\beta_2 = 1.0$	EGMM-logistic	1.029	0.029	0.178	0.176	0.032	0.952
	EGMM-dLASSO	1.028	0.028	0.177	0.176	0.032	0.953
	EGMM-RF	1.037	0.037	0.172	0.176	0.031	0.951
$n = 300; \rho = 0.45$							
$\alpha = 1.0$	EGMM-logistic	1.014	0.014	0.064	0.064	0.004	0.945
	EGMM-dLASSO	1.015	0.015	0.064	0.064	0.004	0.945
	EGMM-RF	1.028	0.028	0.062	0.065	0.005	0.939
$\beta_0 = -0.3$	EGMM-logistic	-0.312	-0.012	0.146	0.142	0.021	0.954
	EGMM-dLASSO	-0.311	-0.011	0.146	0.142	0.021	0.954
	EGMM-RF	-0.293	0.007	0.141	0.141	0.020	0.956
$\beta_1 = -1.0$	EGMM-logistic	-1.021	-0.021	0.105	0.104	0.011	0.947
	EGMM-dLASSO	-1.021	-0.021	0.105	0.104	0.011	0.947
	EGMM-RF	-0.975	0.025	0.102	0.106	0.011	0.946
$\beta_2 = 1.0$	EGMM-logistic	1.034	0.034	0.201	0.201	0.042	0.944
	EGMM-dLASSO	1.034	0.034	0.201	0.201	0.042	0.944
	EGMM-RF	1.033	0.033	0.192	0.197	0.038	0.951

Table 1.4: The estimation of coefficients in timespan part (Continue)

$n = 500; \rho = 0$							
$\alpha = 1.0$	EGMM-logistic	1.008	0.008	0.046	0.046	0.002	0.958
	EGMM-dLASSO	1.008	0.008	0.046	0.046	0.002	0.958
	EGMM-RF	1.012	0.012	0.046	0.047	0.002	0.941
$\beta_0 = -0.3$	EGMM-logistic	-0.308	-0.008	0.109	0.109	0.012	0.949
	EGMM-dLASSO	-0.307	-0.007	0.108	0.109	0.012	0.949
	EGMM-RF	-0.292	0.008	0.105	0.107	0.011	0.941
$\beta_1 = -1.0$	EGMM-logistic	-1.010	-0.010	0.078	0.076	0.006	0.942
	EGMM-dLASSO	-1.009	-0.009	0.078	0.076	0.006	0.946
	EGMM-RF	-0.980	0.020	0.077	0.077	0.006	0.940
$\beta_2 = 1.0$	EGMM-logistic	1.014	0.014	0.138	0.137	0.019	0.951
	EGMM-dLASSO	1.013	0.013	0.138	0.137	0.019	0.951
	EGMM-RF	1.023	0.023	0.134	0.134	0.018	0.948
$n = 500; \rho = 0.45$							
$\alpha = 1.0$	EGMM-logistic	1.009	0.009	0.049	0.049	0.002	0.958
	EGMM-dLASSO	1.009	0.009	0.049	0.049	0.002	0.958
	EGMM-RF	1.022	0.022	0.048	0.049	0.003	0.941
$\beta_0 = -0.3$	EGMM-logistic	-0.309	-0.009	0.109	0.110	0.012	0.954
	EGMM-dLASSO	-0.309	-0.009	0.109	0.110	0.012	0.954
	EGMM-RF	-0.292	0.008	0.106	0.107	0.011	0.944
$\beta_1 = -1.0$	EGMM-logistic	-1.011	-0.011	0.082	0.081	0.007	0.949
	EGMM-dLASSO	-1.011	-0.011	0.082	0.081	0.007	0.947
	EGMM-RF	-0.964	0.036	0.080	0.081	0.008	0.920
$\beta_2 = 1.0$	EGMM-logistic	1.018	0.018	0.155	0.156	0.024	0.947
	EGMM-dLASSO	1.018	0.018	0.155	0.156	0.024	0.947
	EGMM-RF	1.017	0.017	0.148	0.150	0.022	0.945

1.4 Analysis of the online banking data

An anonymous bank in Taiwan provided the sampling data. The total number of customers' samples is 16,588. The data variables include the date of applying for online banking, the date of last use online banking, the customer's age, whether they have a securities account, and the area where the customer is located. The securities account of online banking is for stock and futures exchange and financial management of customer. The customer's age is calculated to 2021. According to the data, the earliest date that the customer applied for

online banking was October 2007. For censoring date, we set the deadline for the last use date of online banking to March 31, 2021. If the used date didn't been recorded, the customer's use time of online banking is denoted censored. We find that the censored percentage is 15.6%. The number of complete observed data is 14000 (84.4%); the number of censored data is 2588 (15.6%). If the date of last use online banking is not recorded in the bank, it means the willingness of using online banking is low. Because the willingness is unobservable, we consider [the modified logistic-Weibull model, modified debiased LASSO-Weibull model and modified random forest Weibull model to analyze this dataset](#). In addition, this dataset is only recorded the last use date of online banking.

Table 1.5 shows the descriptive statistics of explanatory variables in the real data. The average age of customers for online banking is 41.54 years old. The ages of most customer are between 20 and 60 years old, and the age class belongs to the adult and supports the social-economic type. At the same time, 32.11% of the customers who apply for online banking have securities accounts. We are well aware that using online banking is essential to respond to market changes quickly in securities trading.

We would like to understand whether the customers have a securities account will affect the willingness to use online banking, and shorten the timespan of using online banking. We divided the customer's residence area into Yunjianan and non-Yunjianan area, where Yunjianan area includes Yunlin County, Chiayi County, and Tainan City. The bank's main service range is the Yunjianan area; 61.59% of the customers live in the Yunjianan area. Therefore, the interesting issue is whether customers who live in the Yunjianan area will affect the willingness and the timespan of using online banking. Table 1.5 shows the customers who used online banking before the research period end have higher proportions in "owning a securities account" and "living in the Yunjianan area".

Figure 1.1 shows the raw proportions of using online banking for customers of different ages by "whether owning a securities account" and by "whether living in the Yunjianan area". There is a clear crossover trend in holding a securities account, especially between the ages 25 and 60. When customers with ages less than 50 years old, the proportion of using online banking for customers with a securities account is higher than that of customers without a securities account.

But the trend for using online banking is opposite to the ages over 50 years old. In terms of the residential area, when customers were between 25 and 70 years old, there is a higher proportion of using online banking for living in the Yunjianan area than that of non-Yunjianan area. Since the overall trends fluctuate and overlap, we consider the correlations between covariate variables to avoid multicollinearity.

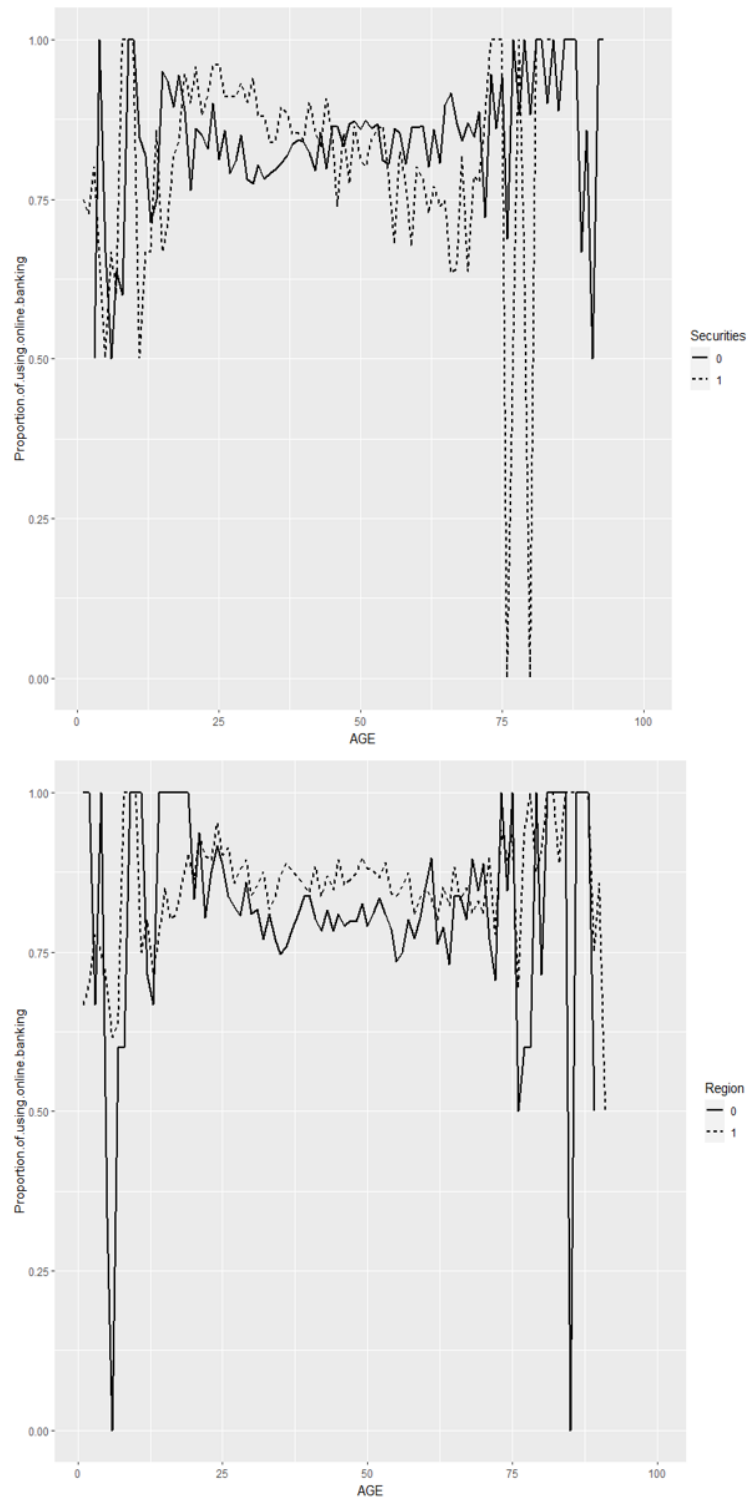


Figure 1.1: The proportion of using online banking before the research end for the customers of different ages by securities account (upper); by the Yunjianan area (lower)

Table 1.5: The descriptive statistics of covariate variables in real data

Variable	Customers of online banking (n=16588)	Uncensored customers (n=14000)	Censored customers (n=2588)
Age(years)	Mean: 41.63	Mean: 41.54	Mean: 42.09
	SD: 13.63	SD: 13.68	SD: 13.32
	Medium:41	Medium: 41	Medium: 41
securities account	Yes(1): 5189(31.28%)	Yes(1): 4496(32.11%)	Yes(1): 693(26.78%)
	No(0): 11399(68.72%)	No(0): 9504(67.89%)	No(0): 1895(73.22%)
live in the Yunjianan area	Yes(1): 10216(61.59%)	Yes(1): 8849(63.21%)	Yes(1): 1367(52.81%)
	No(0): 6372(38.41%)	No(0): 5151(36.79%)	No(0): 1221(47.18%)

Before modeling the real case, we need to check whether "owning a securities account" and "living in the Yunjianan area" are related. From the cross table in Table 1.6, the p-value of the chi-square test is less than the significance level of 0.05. It indicates the two variables are related. It shows that the proportion of owning a securities account for customers who live in the non-Yunjianan area is higher than the customers who live in the Yunjianan area. We can use the Cramer's V coefficient to represent the correlation between "owning a securities account" and "living in the Yunjianan area". It is a numerical indicator to measure the degree of correlation between the two categorical variables. The formula is as follows:

$$V = \sqrt{\frac{\chi^2}{[n \times \min(k - 1, c - 1)]}},$$

where χ^2 is the value of chi-square test statistic, and k and c are the numbers of categories for the two categorical variables, respectively. Because Cramer's V coefficient is 0.076, which means that the correlation between "owning a securities account" and "living in the Yunjianan area" is weak.

As can be seen in Table 1.6, there is a statistically significant difference between the average ages of the two groups, that is, the average age of customers owning an account is lower than that of those who do not. To calculate the correlation between categorical variable and continuous variable, we use the point biserial correlation coefficient r , a correlation coefficient derived from the Pearson correlation coefficient. The measurement of point biserial correlation coefficient calculates the relationship between a binary variable and a continuous variable. The range is also the same as the Pearson correlation coefficient, which is between -1 and 1. Let Y be a continuous variable, $\bar{y}_1$ be the sample mean of Y corresponding to category 1, $\bar{y}_0$ be the sample mean of Y corresponding to category 0; p be the proportion of category 1, q be the proportion of category 0; S_y be the sample standard deviation of Y .

The point biserial correlation coefficient is:

$$r = \frac{(\bar{y}_1 - \bar{y}_0)\sqrt{pq}}{S_y}.$$

The correlation between the age and owning a securities account is statistically significant, with $r = -0.226$. Therefore, we added the interaction term of owning a securities account and the age of the customers into [three modified mixture models](#). In addition, Table 1.6 shows that there is a statistically significant difference between the average ages of customers residing in the Yunjianan area and the non-Yunjianan area. The average age of customers living in the Yunjianan area is higher than those not in the Yunjianan area. We calculated the point biserial correlation coefficient of the age and living in the Yunjianan area, and the result was $r = 0.05$. Because the correlation is not strong, the interaction term "living in the Yunjianan area" and "age" was not added to [three modified mixture models](#). Then, we fit the modified logistic regression-Weibull model, the modified debiased LASSO-Weibull model and the modified random forest-Weibull model.

Table 1.6: The descriptive statistics of covariate variables in real data

		owning a securities account		Chi-square test	Cramer's V
		Yes	No	<i>p value</i>	
living in the Yunjianan area	Yes	2910(29.48%)	7306(71.52%)	< 0.001	0.076
	No	2279(35.77%)	4093(64.23%)		
		Mean age	SD of age	Mann-Whitney U test	Point biserial correlation
owning a securities account	Yes	37.07	13.16	< 0.001	-0.226
	No	43.70	13.33		
Living in the Yunjianan area	Yes	42.22	14.39	< 0.001	0.050
	No	40.68	12.24		

Table 1.7 shows the results of three modified mixture models. From the Akaike information criterion (AIC) viewpoint, the AIC of the modified logistic-Weibull model is 118054.4, the AIC of modified random forest-Weibull model is 110411.4, and the modified debiased LASSO-Weibull model is 143483. The AIC of modified random forest-Weibull model is smallest. Regarding willingness of using online banking, the standard errors of the estimator by employing the modified debiased LASSO-Weibull model is smaller than modified logistic-Weibull model. The securities account and interaction between age and owning a securities account are significant for two models, while age and residential area are insignificant for the modified logistic-Weibull model. However, the hypothesis testing for modified logistic-Weibull model is different to the modified debiased LASSO-Weibull model in residential

area. Because the estimated variance of coefficient estimator of living in the Yunjianan area might be inflated in the modified logistic-Weibull model, the result of the mixture debiased LASSO model is worthy of adoption. Customers who live in non-Yunjianan area and own a securities account are more likely to be willing to use online banking.

In a scenario where customers have a willingness to use online banking, the results presented in Table 1.8 indicate a decreasing trend in the hazard ratio (HR) estimates across three models as age increases. This suggests that, with a consistent consideration of variables such as "Securities account" and "Residential area," the time span between instances of utilizing online banking tends to be shorter for younger individuals.

Moreover, at a fixed age, the HR estimates are positive for customers who own a securities account compared to those who do not, as indicated by the three modified mixture models. This implies that having a securities account is associated with a shortened time span between instances of using online banking. Similarly, the HR estimates are positive for customers residing in the Yunjianan area compared to those who do not, according to the results from three models. This suggests that living in the Yunjianan area is associated with a shortened time span between instances of utilizing online banking.

Judging from the absolute value of the coefficient, the impact of "owning a securities account" is more significant than the impact of "living in the Yunjianan area". When the age of customers is fixed at 41 years old, the median of the estimated timespan of online account use for customers with using willingness is shown in Table 1.8.

Table 1.7: Modified debiased LASSO-Weibull model and modified random forest-Weibull model

The willingness of using online banking												
	Intercept		Age		securities account		Residential area		Interaction			
	Logit	dLASSO	Logit	dLASSO	Logit	dLASSO	Logit	dLASSO	Logit	dLASSO		
Coef.	2.521	3.373	0.023	0.066	4.362	4.156	0.883	-0.688	-0.097	-0.139		
SE	1.084	0.301	0.014	0.006	1.211	0.360	0.721	0.212	0.017	0.008		
OR			1.024	1.068	78.396	63.828	2.419	0.503	0.907	0.870		
p value	0.021	0.001*	0.094	0.001*	0.001*	0.001*	0.220	0.001	0.001*	0.001*		
The timespan of using online banking												
	Intercept			Age			securities account			Residential area		
	Logit	dLASSO	RF	Logit	dLASSO	RF	Logit	dLASSO	RF	Logit	dLASSO	RF
Coef.	-2.056	-3.196	-2.636	-0.001	-0.002	-0.002	0.478	0.743	0.632	0.231	0.318	0.260
SE	0.055	0.037	0.052	0.001	0.001	0.001	0.030	0.020	0.019	0.029	0.015	0.013
HR				0.999	0.998	0.998	1.613	2.102	1.881	1.260	1.374	1.297
p value	0.001*	0.001*	0.001*	0.072	0.001*	0.001*	0.001*	0.001*	0.001*	0.001*	0.001*	0.001*
0.001* means < 0.001.												

0.001* means < 0.001 .

Table 1.8: The median of the estimated timespans for three modified mixture models for willing to use online banking

		owning a securities account		
			Yes	No
Modified logistic-Weibull model	Living in theYunjianan area	Yes	26.10	118.21
		No	54.11	245.08
Modified debiaed LASSO-Weibull model	Living in theYunjianan area	Yes	63.82	339.42
		No	130.35	693.29
Modified random forest-Weibull model	Living in theYunjianan area	Yes	37.64	183.50
		No	72.24	352.16

1.5 Conclusions and discussions

Except for modified logistic-Weibull model, we introduce the modified debiased LASSO-Weibull model and the modified random forest-Weibull model to investigate customers' inclination and the duration of online banking usage, categorized by specific traits. Analyzing a real dataset reveals that factors such as "customer's age," "ownership of a securities account," and "residential area" significantly influence both the willingness and the duration of online banking usage. Identifying these variables can assist banks in precisely targeting customer promotions, thereby reducing marketing costs associated with customers less likely to engage with online resources or having a shorter duration of engagement.

The modified logistic-Weibull model and the modified debiased LASSO-Weibull model are parametric methods, while the modified random forest-Weibull model is nonparametric. Each approach has its strengths and limitations. Optimal model selection depends on the nature of the data. For nonlinear relationships between the response and covariates, the modified random forest-Weibull model proves more suitable, though consistency of preferences estimated by random forest requires verification. Based on the findings from a real-case analysis, when examining the association between the response and covariates using a logit form, the modified debiased LASSO-Weibull model demonstrates superior performance in variance estimation compared to the modified logistic-Weibull model. On the other hand, expert opinions and banking experience were also taken into account when choosing the appropriate model, aiming to enhance overall model performance. Future iterations of the models will incorporate additional relevant independent variables, enabling a comprehensive estimation of customer preferences and the corresponding duration of online banking usage.

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Disclosure statement

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References

- Anatolyev, S., GMM, GEL, serial correlation, and asymptotic bias. *Econometrica*, 2005, 73, 983-1002.
- Annin, K., Omane-Adjepong, M., Senya, S.S., Applying logistic regression to e-banking usage in Kumasi Metropolis, Ghana. *International Journal of Marketing Studies*, 2014, 6, 153-162.
- Berkson, J., Gage, R.P., Survival curve for cancer patients following treatment. *Journal of the American Statistical Association*, 1952, 47.259, 501-515.
- Boag, J.W., Maximum likelihood estimates of the proportion of patients cured by cancer therapy. *Journal of the Royal Statistical Society. Series B (Methodological)*, 1949, 11.1, 15-53.
- Breiman, L., Random forests. *Machine Learning*, 2001, 45.1, 5-32.
- Calisir, F, Gumussoy, CA, Internet banking versus other banking channels: Young consumers' view. *International Journal of Information Management*, 2008, 28, 215-221.
- Carrasco, M, A regularization approach to the many instruments problem, *Journal of Econometrics*, 2012, 170, 383-398.
- Chaussé, P, Computing generalized method of moments and generalized empirical likelihood with R, *Journal of Statistical Software*, 2010, 34, 1-35.
- Chen, C.H., Tsay, Y.C., Wu, Y.C., Horng, C.F., Logistic-AFT location-scale mixture regression models with nonsusceptibility for left-truncated and general interval-censored data. *Statistics in Medicine*, 2013, 32.24, 4285-4305.
- Chen, I.C., Westgate, P.M., Improved methods for the marginal analysis of longitudinal data in the presence of time-dependent covariates. *Statistics in Medicine*, 2017, 36.16, 2533-2546.

- Chen, I.C., Westgate, P.M., A novel approach to selecting classification types for time-dependent covariates in the marginal analysis of longitudinal data. *Statistical Methods in Medical Research*, 2019, 28, 3176-3186.
- Chen, Y.R., Su, F., Zhang, R.Y., Discussion on the management strategies of domestic online banking. *Taiwan Financial Quarterly*, 2002, 3, 1-19. (in Chinese)
- Crede, A, Electronic commerce and the banking industry: the requirement and opportunities for new payment systems using the Internet. *Journal of Computer Mediated Communication*, 1995, 1, JCMC135.
- Corbiere, F., Joly, P., A SAS macro for parametric and semiparametric mixture cure models. *Computer Methods and Programs in Biomedicine*, 2007, 85.2, 173-180.
- Cortes, C, Vapnik, V., Support-vector networks. *Machine Learning*, 1995, 20.3, 273-297.
- Dempster, A.P., Laird, N.M., and Rubin, D.B. Maximum likelihood from incomplete data via the EM algorithm. *Journal of the Royal Statistical Society. Series B*, 1977, 39, 1-38.
- Diggle, P.J., Heagerty, P., Liang, K.Y., Zeger, S.L., *Analysis of longitudinal data*. Oxford University Press, 2002.
- Duasa, J., Nazri, N.J.Z., Zainal, N.H., Likelihood of using online banking services among consumers: Application of the logit model. *Malaysian Journal of Consumer and Family Economics*, 2019, 22, 220-232.
- Engelhard, M., Henao, R., Disentangling Whether from When in a Neural Mixture Cure Model for Failure Time Data. *International Conference on Artificial Intelligence and Statistics*, 2022, 9571-9581.
- Efron, B., Nonparametric estimates of standard error: the jackknife, the bootstrap and other methods. *Biometrika*, 1981, 68.3: 589-599.
- Fan, J., Li, R., Variable selection for Cox's proportional hazards model and frailty model. *The Annals of Statistics*, 2002, 30.1, 74-99.
- Farewell, V.T., A model for a binary variable with time-censored observations. *Biometrika*, 1977, 64.1, 43-46.
- Farewell, V.T., The use of mixture models for the analysis of survival data with long-term survivors. *Biometrics*, 1982, 1041-1046.

- Fu, H., Nicolet, D., Mrózek, K., Stone, R.M., Eisfeld, A. K., Byrd, J. C., Archer, K.J., Controlled variable selection in Weibull mixture cure models for high-dimensional data. *Statistics in Medicine*, 2022, 41.22, 4340-4366.
- Gamel, J.W., McLean, I.W., Rosenberg, S.H., Proportion cured and mean log survival time as functions of tumor size. *Statistics in Medicine*, 1990, 9, 999-1006.
- Ghitany, M.E., Maller, R.A., Asymptotic results for exponential mixture models with long-term survivors. *Statistics: A Journal of Theoretical and Applied Statistics*, 1992, 23.4, 321-336.
- Goldman, A.I., Survivorship analysis when cure is a possibility: A Monte Carlo study. *Statistics in Medicine*, 1984, 3, 153-163.
- Gordon, N.H., Maximum likelihood estimation for mixtures of two gompertz distributions when censoring occurs. *Communications in Statistics-Simulation and Computation*, 1990, 19, 733-747.
- Hanin, L., Huang, L.S. Identifiability of cure models revisited. *Journal of Multivariate Analysis*, 2014, 130, 261-274.
- Hansen, L.P., Large sample properties of generalized method of moments estimators. *Econometrica: Journal of the Econometric Society*, 1982, 1029-1054.
- Han, P., Song, P.X.K., A note on improving quadratic inference functions using a linear shrinkage approach. *Statistics & Probability Letters*, 2011, 81.3, 438-445.
- Hoerl, A.E., Kennard, R.W., Ridge regression: Biased estimation for nonorthogonal problems. *Technometrics*, 1970, 12, 55-67.
- Jiang, C., Wang, Z., Zhao, H., A prediction-driven mixture cure model and its application in credit scoring. *European Journal of Operational Research*, 2019, 277, 20-31.
- Jones, D.R., Powles, R.L., Machin, D., Sylvester, R.J., On estimating the proportion of cured patients in clinical studies. *Biometrie-Praximetrie*, 1981, 21, 1-11.
- Kaplan, E.L., Meier, P., Nonparametric estimation from incomplete observations. *Journal of the American statistical association*, 1958, 53.282, 457-481.
- Kuan, C.M., Generalized Method of Moment. Taiwan: Department of Finance & CRETA National Taiwan University, 2010.

- Kumari, J.P., A Study of online banking usage among University Academics. *Social Sciences*, 2016, 18, 32-37.
- Larson, M.G., Dinse, G.E., A mixture model for the regression analysis of competing risks data. *Journal of the Royal Statistical Society. Series C (Applied Statistics)* 1985, 34.3, 201-211.
- Lassar, W.M., Manolis, C., Lassar, S.S., The relationship between consumer innovativeness, personal characteristics, and online banking adoption. *International Journal of Bank Marketing*, 2005, 23, 176-199.
- Lai, T. L., Small, D., Marginal regression analysis of longitudinal data with time-dependent covariates: a generalized method-of-moments approach. *Journal of the Royal Statistical Society. Series B (Statistical Methodology)* 2007, 69.1, 79-99.
- Lalonde, T. L., Nguyen, A. Q., Yin, J., Irimata, K., Wilson, J.R., Modeling correlated binary outcomes with time-dependent covariates. *Journal of Data Science*, 2013, 11.4.
- Li, C.S., Taylor, J.M., Sy, J.P., Identifiability of cure models. *Statistics & Probability Letters*, 2001, 54.4, 389-395.
- Li, P., Peng, Y., Jiang, P., Dong, Q., A support vector machine based semiparametric mixture cure model. *Computational Statistics*, 2020, 35.3, 931-945.
- Liang, K.Y., Zeger, S.L., Longitudinal data analysis using generalized linear models. *Biometrika*, 1986, 73.1, 13-22.
- Louis, T.A., Finding the observed information matrix when using the EM algorithm. *Journal of the Royal Statistical Society. Series B (Methodological)* 1982, 44, 226-233.
- Liu, X., Peng, Y., Tu, D., Liang, H., Variable selection in semiparametric cure models based on penalized likelihood, with application to breast cancer clinical trials. *Statistics in Medicine*, 2012, 31.24, 2882-2891.
- Mavri, M., Ioannou, G., Consumers' perspectives on online banking services. *International Journal of Consumer Studies*, 2006, 30, 552-560.
- Mo, L.S., Yau, K.K., Survival mixture model for credit risk analysis. *Asia-Pacific Journal of Risk and Insurance*, 2010, 4.
- Nehmzow, C., The Internet will shake banking's medieval foundations, *Journal of Internet Banking and Commerce*, 1997, 2, 1-2.

- Newey, W.K., Smith, R.J., Higher order properties of GMM and generalized empirical likelihood estimators. *Econometrica*, 2004, 72, 219-255.
- Niu, Y., Peng, Y., Marginal regression analysis of clustered failure time data with a cure fraction. *Journal of Multivariate Analysis*, 2014, 123, 129-142.
- Ooi, S., Surge in E-commerce Transactions. *SME IT Guid*, 1999, 6.
- Platt, J., Sequential minimal optimization: a fast algorithm for training support vector machines. <https://www.microsoft.com/en-us/research/wp-content/uploads/2016/02/tr-98-14.pdf>. Accessed November 3 2019.
- Pal, S., Peng, Y., Barui, S., Wang, P., A Support Vector Machine Based Cure Rate Model For Interval Censored Data. arXiv preprint arXiv:2109.01098, 2021.
- Platt, J., Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods. *Advances in Large Margin Classifiers*, 1999, 10.3, 61-73.
- Peng, Y., Fitting semiparametric cure models. *Computational Statistics & Data Analysis*, 2003, 41, 481-490.
- Peng, Y., Dear, K.B., Denham, J.W., A generalized F mixture model for cure rate estimation. *Statistics in Medicine*, 1998, 17.8, 813-830.
- Peng, Y., Dear, K.B., A nonparametric mixture model for cure rate estimation. *Biometrics*, 2000, 56.1, 237-243.
- Pepe, M.S., Anderson, G.L., A cautionary note on inference for marginal regression models with longitudinal data and general correlated response data. *Communications in Statistics-Simulation and Computation*, 1994, 23.4, 939-951.
- Therneau, T., Grambsch, P., *Modeling Survival Data: Extending the Cox Model*. Springer-Verlag, New York. ISBN: 0-387-98784-3, 2000.
- Tibshirani, R., Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society. Series B (Methodological)*, 1996, 58, 267-288.
- Tong, E.N., Mues, C., Thomas, L.C., Mixture cure models in credit scoring: If and when borrowers default. *European Journal of Operational Research*, 2012, 218, 132-139.
- Qu, A., Lindsay, B.G., Li, B., Improving generalised estimating equations using quadratic inference functions. *Biometrika*, 2000, 87.4, 823-836.

- Rosen, O., Jiang, W., Tanner, M.A., Mixtures of marginal models. *Biometrika*, 2000, 87.2, 391-404.
- Van de Geer, S., Bühlmann, P., Ritov, Y.A., Dezeure, R., On asymptotically optimal confidence regions and tests for high-dimensional models, *The Annals of Statistics*, 2014, 42, 1166-1202.
- Wolter, M., Rösch, D., Cure events in default prediction. *European Journal of Operational Research*, 2014, 238, 846-857.
- Yamaguchi, K., Accelerated failure-time regression models with a regression model of surviving fraction: an application to the analysis of permanent employment in Japan. *Journal of the American Statistical Association*, 1992, 87.418, 284-292.
- Yang, Y.H., Ma, M.C. A Comparison on Different Mixed Cured Model with the Cox Proportional Hazard Function. Master thesis, National Cheng Kung University, Taiwan, ROC, 2017.
- Yu, B., Peng, Y., Mixture cure models for multivariate survival data. *Computational Statistics & Data Analysis*, 2008, 52.3, 1524-1532.
- Yin, J., Correlated GMM Logistic Regression Models with Time-Dependent Covariates and Valid Estimating Equations. Arizona State University, 2012.
- Zadrozny, B., Elkan, C., Transforming classifier scores into accurate multiclass probability estimates. In *Proceedings of the eighth ACM SIGKDD international conference on Knowledge discovery and data mining*, 2002, 694-699.
- Zhou, Y., Lefante, J., Rice, J., Chen, S., Using modified approaches on marginal regression analysis of longitudinal data with time-dependent covariates. *Statistics in Medicine*, 2014, 33.19, 3354-3364.
- Zhu, X., High-dimensional Bias-corrected Inference, with Applications to fMRI Studies. Doctoral dissertation, University of Pittsburgh, 2019.